STEVEN KOLAWOLE

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RESEARCH INTEREST

Inference efficiency for LLMs: developing methods for compression, resource optimization (KV cache), adaptive serving (cascading, speculative decoding), and inference parallelism for production systems.

EDUCATION

Ph.D., Carnegie Mellon University, USA — 2023 - 2028
Language Technologies Institute, School of Computer Science

BSc., Federal University of Agriculture Abeokuta, Nigeria — 2017 - 2023
Department of Computer Science

RESEARCH EXPERIENCE

AWS Bedrock, Inference Optimization Science Team — San Francisco, CA
Applied Scientist Intern — *May - Aug 2026*
Mentored by Drs Nathan Pemberton & Kyle Ulrich. Focus: compression + parallel token prediction to maximize the compute/latency efficiency of disaggregated speculative decoding setups.

Carnegie Mellon University — Pittsburgh, PA
Graduate Student Researcher — *Aug 2023 - Present*
Advised by Professor Virginia Smith. Developing algorithmic methods for efficient ML inference.

ML Collective — remote
Independent Researcher — *Apr 2021 - Present*
Pre-PhD work mentored by Drs Rosanne Liu & Jason Yosinski; now mentoring junior researchers within the same open-science collab pipeline. Research focus: resource-constrained NLP, federated learning, and efficient ML.

German Research Centre for Artificial Intelligence (DFKI) — remote
Undergraduate Research Assistant with Nils Rethmeier — *Fall 2021*
Investigated resource-efficient models (CLESS) vs. large, self-supervised models on fine-grained downstream tasks.

PUBLICATIONS

[7] **Steven Kolawole** and Virginia Smith. "Epiphany-Aware KV Cache Eviction Without the Attention Matrix." [EMNLP 2026]

[6] Pearse Jim*, **Steven Kolawole***, Opegbemi M. Busoye, Glory Bagai, and Virginia Smith. "What Survives When You Compress a Recursive Reasoner for the Edge?" [EMNLP 2026]

[5] **Steven Kolawole**, Keshav Santhanam, Virginia Smith, and Pratiksha Thaker. "ParallelPrompt: Extracting Parallelism from Large Language Model Queries." [NeurIPS 2025]

[4] Duncan Soiffer, **Steven Kolawole**, and Virginia Smith. "Semantic Agreement Enables Efficient Open-Ended LLM Cascades." [EMNLP 2025]

[3] **Steven Kolawole***, Don Dennis*, Ameet Talwalkar, and Virginia Smith. "Agreement-Based Cascading for Efficient Inference." [TMLR 2025]

[2] **Steven Kolawole***, Lucio Dery*, JF Kagy, Virginia Smith, Graham Neubig, and Ameet Talwalkar. "Everybody Prune Now: Structured Pruning of LLMs with only Forward Passes." [arXiv preprint]

[1] **Steven Kolawole**, Opeyemi Osakuade, Nayan Saxena, and Babatunde Kazeem Olorisade. "Sign-to-Speech Model for Sign Language Understanding: A Case Study of Nigerian Sign Language." [IJCAI 2022]

WORKSHOP PAPERS

[5] Mardiyyah Oduwole, ..., Abraham Owodunni, and **Steven Kolawole**. "Scarcity to Efficiency: Investigating the Effects of Data Augmentation on African Machine Translation." [LM4UC, AAAI '26 (Springer CCIS)]

[4] Nnaemeka Obiefuna*, Samuel Oyeneye*, Similoluwa Odunaiya*, Iremide Oyelaja*, and **Steven Kolawole**. "PrivacyBench: Privacy Isn't Free in Hybrid Privacy-Preserving Vision Systems." [ES-FoMo, ICML '25]

[3] Busayo Awobade*, Mardiyyah Oduwole*, and **Steven Kolawole***. "What Happens When Small Is Made Smaller? Exploring the Impact of Compression on Small-Data Language Models." [AfricaNLP, ICLR '24]

[2] Colin Leong, ..., **Steven Kolawole** et al. "Adapting to the Low-Resource Double-Bind: Investigating Low-Compute Methods on Low-Resource African Languages." [AfricaNLP, ICLR '23]

[1] Nahid Alam*, **Steven Kolawole***, Simardeep Sethi*, Nishant Bansali, and Karina Nguyen. "Vision Transformers for Mobile Applications: A Short Survey." [arXiv preprint '23]

* equal contribution.

SELECTED HONORS & AWARDS

Scholar, MLSS NYC Competitively selected as one of ∼200 global PhD researchers for the flagship ML summer school; awarded full travel funding. Curriculum covered: Agentic AI, LLM efficiency, alignment, and safety. **(2026)**

Grant Recipient, Algorand Foundation Awarded $115K grant to develop the ASAlytics platform for real-time opinion mining of digital assets. **(2022)**

Sign Language Research Awards {National AI Champion, Nigeria Computer Society (2022); Ideathon Winner ($10K GCP credits), Deep Learning Indaba (2022); Best Poster, Data Science Nigeria (2021); Winner, DeepQuest AI Challenge (2021)} — multiple recognitions for research bridging gaps for sub-Saharan Africa's hearing impaired.

Scholar, MTN Foundation Science & Technology Scholarship Academic excellence recognition for high-performing, low-income students in Nigerian public institutions. *Selection rate: 1.8%* **(2020, 2021)**

Mr. Algorithm (1st Runner-up) Data Science Nigeria's recognition for outstanding contributions to Nigeria's AI ecosystem via technical excellence and knowledge sharing. **(2020)**

Gold Medal Winner, NaijaHacks '19 One of the two AgroAI team members that built a mobile application that helps farmers detect plant diseases using Computer Vision. *Rank:* top 12 / 336 teams at the final Expo day. **(2019)**

SELECTED TALKS & PRESENTATIONS

IJCAI '22 & NeurIPS ML4D '21 Oral presentations of accepted paper "Sign-to-Speech Model for Sign Language Understanding: A Case Study of Nigerian Sign Language."

Black in AI's ELAI Program & CMU Africa's Research Club Repeated guest speaker on graduate school applications and research career development. (2024-2025)

Cohere For AI Independent Research Panel Panel discussion on thriving as an independent researcher and securing research collaborations. (2023)

SELECTED COMMUNITY IMPACT & SERVICE

Founding Organizer, ML Collective-Africa Facilitating a grassroots research hub, mentoring URM/constrained students to develop independent research skills via P2P study, collaborative projects, and resource support. (2023–)

Mentor, STEM for Development Coaching URM graduate school aspirants to clarify research interests and optimize applications for Western graduate programs. Currently mentoring 20+ students. (2023–2026)

Community Lead, Google Developer Students Club, FUNAAB Organized training impacting 3000+ students, personally teaching 600+ students in ML, data science, and technical skills. (2021–2022)

Campus Lead, AI+ FUNAAB Transformed club into a top-3 Nigerian student AI community, enabling record African school undergraduate participations at Deep Learning Indaba (9 in '22; 23 students in '23). (2019–2021)

SELECTED WORK EXPERIENCE

Founding ML Engineer 2022

ASAlytics (now Nazari)

- Built backend systems for opinion mining/analytics for assets on Algorand blockchain via social media scraping;
- Developed scalable data pipeline handling social media feeds and sentiment analysis for cryptocurrency markets.

Data Science Intern 2021

SeqHub Analytics LLC *New Haven, CT*

- Fine-tuned mT5 for machine translation on low-resource Nigerian languages;
- Built microservices for conversational AI agent including edit prediction functionality using Levenshtein Distance;
- Built dockerized dashboard automating analytics workflow, boosting development speed by 45%.